

Exercise 1 [7 pt] Consider E any normed vectoriel space and E' its topological dual.

- for $A \subset E$ we define $A^\perp = \{f \in E', f(x) = 0, \forall x \in A\}$. Show that
 - If $A_1 \subset A_2$, then $A_2^\perp \subset A_1^\perp$,
 - $A^\perp = [\text{vect}(A)]^\perp$,
 - $A^\perp$ is a closed linear sub-space of E' and $E^\perp = \{0\}$.
- For $B \subset E'$, we define ${}^\perp B = \{x \in E, f(x) = 0, \forall f \in B\}$. Show that
 - If $B_1 \subset B_2$, then ${}^\perp B_2 \subset {}^\perp B_1$,
 - ${}^\perp B = {}^\perp [\text{vect}(B)]$,
 - ${}^\perp B$ is a closed linear sub-space of E ,
 - For any $A \subset E$ and any $B \subset E'$ we have $\bar{A} \subset {}^\perp (A^\perp)$ and $\bar{B} \subset ({}^\perp B)^\perp$.

Exercise 2 [7 pt] Let us consider the space $E = C([0, 1], \mathbb{R})$ of continuous functions defined on the interval $[0, 1]$, equipped with the norm $\|\cdot\|_\infty$.

- Show that $(E, \|\cdot\|_\infty)$ is not a Hilbert space. (Hint: Take $f(x)=x$ and $g(x)=1-x$)
- Show that $(E, \|\cdot\|_\infty)$ is complete.
- Consider the set $F = \{f \in E : f(0) = 0 \text{ and } \int_0^1 f(x)dx \geq 1\}$.
 - Show that F is closed and convex. (If $f, g \in F$ and $\lambda \in [0, 1]$ then $\lambda f + (1-\lambda)g \in F$).
 - Show that for any $f \in F$, we have $\|f\|_\infty > 1$. (Suppose to the contrary that there exists $f_0 \in F$ such that $\|f_0\|_\infty \leq 1$. Consider the cases $\|f_0\|_\infty = 1$ and $\|f_0\|_\infty < 1$)
 - In the case $\|f_0\|_\infty = 1$, considere the set $A = \{x : f(x) < 1\}$.
 - Show that $\inf_{f \in F} \|f\| = 1$. (Take a sequence f_n , such that $f_n \in F$ and $\lim \|f_n\| = 1$).
 - Conclude

Exercise 3 [6 pt] Let H be a Hilbert space, (A_n) a non increasing $(A_{n+1} \subset A_n)$ sequence of non empty bounded closed **convex** subsets of H and let $x_0 \in H, x_0 \notin A_1$. Put a_n the unique point in A_n such that $\|x_0 - a_n\| = d(x_0, A_n)$, define $\alpha_n = d(x_0, A_n)$

- Prove that if $n < m$ then $\alpha_n \leq \alpha_m$,
- Show that the sequence α_n is bounded and deduce that it is convergente,
- Show that

$$\frac{1}{2} \|a_n - a_m\|^2 = \|x_0 - a_n\|^2 + \|x_0 - a_m\|^2 - 2 \left\| x_0 - \frac{a_n + a_m}{2} \right\|^2$$

- Show that the sequence a_n is a Cauchy sequence.
- Deduce that $\bigcap_{k \geq 1} A_k \neq \emptyset$.



Good Luck

lieve that mathematics is simple, it is only because they do not realize how complicated

John von Neumann